

BANK Database – DQL & DML Tasks

DQL

1. Display all customer records.

SQLQuery college.s...BTESAM\ebtes (79))* SQLQuery_CODING....TESAM\ebtes (65))*
SQLQuery_CODING....TESAM\ebtes (99))*

```
-----solve DQL -----  
--1--  
SELECT *  
FROM Customer_table1;
```

100 %

Results Messages

	customer_id	name	address	phone_number	date_of_birth
1	1	ebtesam	123 bebo, Muscat	977001122	1998-04-12
2	2	Fatma	45 Sohar St, Sohar	977334455	1992-08-23
3	3	Omar	67 Nizwa Rd, Nizwa	977556677	1978-11-05
4	4	Amani	89 swaiq Ave, albatinh	977778899	1989-01-20

2. Display customer full name, phone, and membership start date.

SQLQuery college.s...BTESAM\ebtes (79))* SQLQuery_CODING....TESAM\ebtes (65))*
SQLQuery_CODING....TESAM\ebtes (99))*

```
--2--  
ALTER TABLE Customer_table1  
ADD membership_start_date DATE;  
  
UPDATE Customer_table1  
SET membership_start_date =  
CASE  
    WHEN customer_id = 1 THEN '2022-01-10'  
    WHEN customer_id = 2 THEN '2023-03-15'  
    WHEN customer_id = 3 THEN '2021-07-20'  
    WHEN customer_id = 4 THEN '2024-05-30'  
    ELSE NULL  
END;  
  
SELECT  
    name AS FullName,  
    phone_number AS Phone,  
    membership_start_date AS MembershipStartDate  
FROM Customer_table1;
```

100 %

Results Messages

	FullName	Phone	MembershipStartDate
1	ebtesam	977001122	NULL
2	Fatma	977334455	NULL
3	Omar	977556677	NULL
4	Amani	977778899	NULL

3. Display each loan ID, amount, and type.

The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains the following SQL code:

```
SELECT
    loan_id AS LoanID,
    amount AS LoanAmount,
    loan_type AS LoanType
FROM Loan_table;
```

The results pane displays the following data:

	LoanID	LoanAmount	LoanType
1	1	25000.00	Home Loan
2	2	10000.00	Car Loan
3	3	5000.00	Personal Loan

4. Display account number and annual interest (5% of balance) as AnnualInterest.

The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains the following SQL code:

```
SELECT
    account_number AS AccountNumber,
    balance * 0.05 AS AnnualInterest
FROM Account_table;
```

The results pane displays the following data:

	AccountNumber	AnnualInterest
1	1	250.0000
2	2	60.0000
3	3	400.0000
4	4	15.0000
5	5	750.0000

5. List customers with loan amounts greater than 100000 LE.

The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains the following SQL code:

```
---5---  
SELECT  
    c.customer_id,  
    c.name AS FullName,  
    l.amount AS LoanAmount,  
    l.loan_type AS LoanType  
FROM Customer_table1 c
```

The results pane shows a table with 5 columns: customer_id, FullName, LoanAmount, and LoanType. The table contains 3 rows of data:

	customer_id	FullName	LoanAmount	LoanType
1	1	ebtesam	25000.00	Home Loan
2	2	Fatma	10000.00	Car Loan
3	4	Amani	5000.00	Personal Loan

6. List accounts with balances above 20000.

The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains the following SQL code:

```
---6---  
SELECT  
    account_number AS AccountNumber,  
    balance AS Balance  
FROM Account_table  
WHERE balance > 2000;
```

The results pane shows a table with 2 columns: AccountNumber and Balance. The table contains 3 rows of data:

	AccountNumber	Balance
1	1	5000.00
2	3	8000.00
3	5	15000.00

7. Display accounts opened in 2023.

The screenshot shows the SQL Server Enterprise Manager interface. The query editor displays the following SQL query:

```
---7---  
SELECT  
    account_number AS AccountNumber,  
    date_creation AS DateCreated,  
    balance,  
    type  
FROM Account_table  
WHERE YEAR(date_creation) = 2023;
```

The Results pane shows the following data:

	AccountNumber	DateCreated	balance	type
1	1	2023-01-15	5000.00	savings
2	5	2023-11-11	15000.00	savings

8. Display accounts ordered by balance descending.

The screenshot shows the SQL Server Enterprise Manager interface. The query editor displays the following SQL query:

```
---8---  
SELECT  
    account_number AS AccountNumber,  
    balance AS Balance,  
    type AS AccountType,  
    date_creation AS DateCreated  
FROM Account_table  
ORDER BY balance DESC;
```

The Results pane shows the following data:

	AccountNumber	Balance	AccountType	DateCreated
1	5	15000.00	savings	2023-11-11
2	3	8000.00	savings	2022-03-20
3	1	5000.00	savings	2023-01-15
4	2	1200.00	checking	2024-07-10
5	4	300.00	checking	2025-02-05

9. Display the maximum, minimum, and average account balance.

The screenshot shows the SQL Server Enterprise Manager interface. The top pane displays the query: `SELECT MAX(balance) AS MaxBalance, MIN(balance) AS MinBalance, AVG(balance) AS AvgBalance FROM Account_table;`. The bottom pane shows the results of the query in a table with three columns: MaxBalance, MinBalance, and AvgBalance. The first row of data shows values 15000.00, 300.00, and 5900.000000 respectively.

```
SQLQuery college.s...BTESAM\ebtes (79))*  SQLQuery_CODING....TESAM\ebtes (65))*  SQLQuery_CODING....TESAM\ebtes (99))* X
```

---9---

```
SELECT
    MAX(balance) AS MaxBalance,
    MIN(balance) AS MinBalance,
    AVG(balance) AS AvgBalance
FROM Account_table;
```

100 %

Results Messages

	MaxBalance	MinBalance	AvgBalance
1	15000.00	300.00	5900.000000

10. Display total number of customers.

The screenshot shows the SQL Server Enterprise Manager interface. The top pane displays the query: `SELECT COUNT(*) AS TotalCustomers FROM Customer_table1;`. The bottom pane shows the results of the query in a table with one column: TotalCustomers. The first row of data shows the value 4.

```
SQLQuery college.s...BTESAM\ebtes (79))*  SQLQuery_CODING....TESAM\ebtes (65))*  SQLQuery_CODING....TESAM\ebtes (99))* X
```

---10---

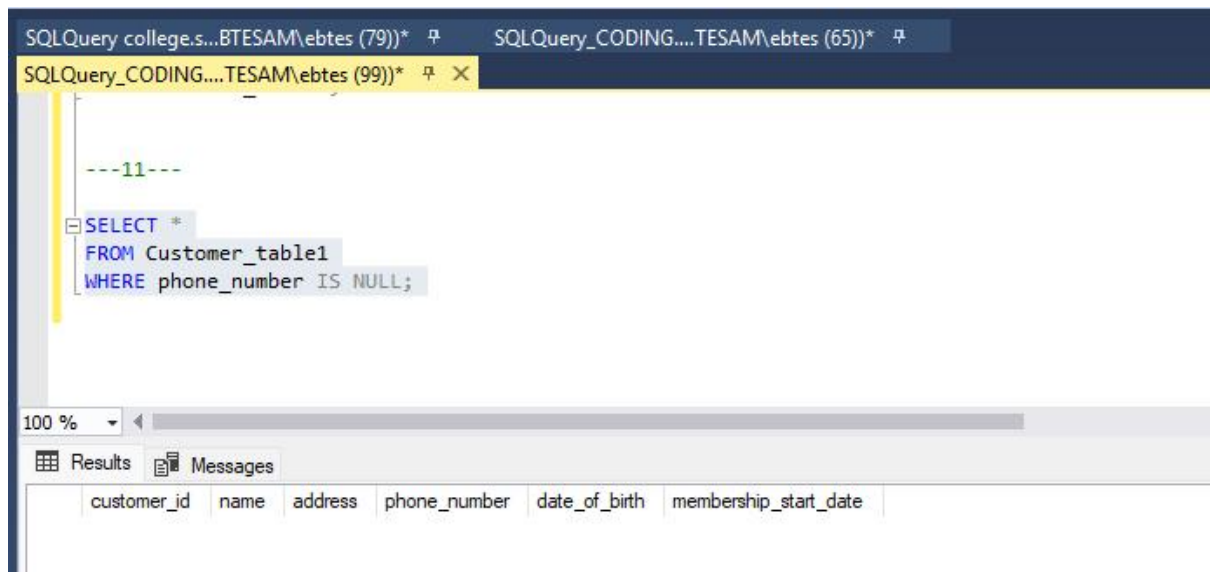
```
SELECT COUNT(*) AS TotalCustomers
FROM Customer_table1;
```

100 %

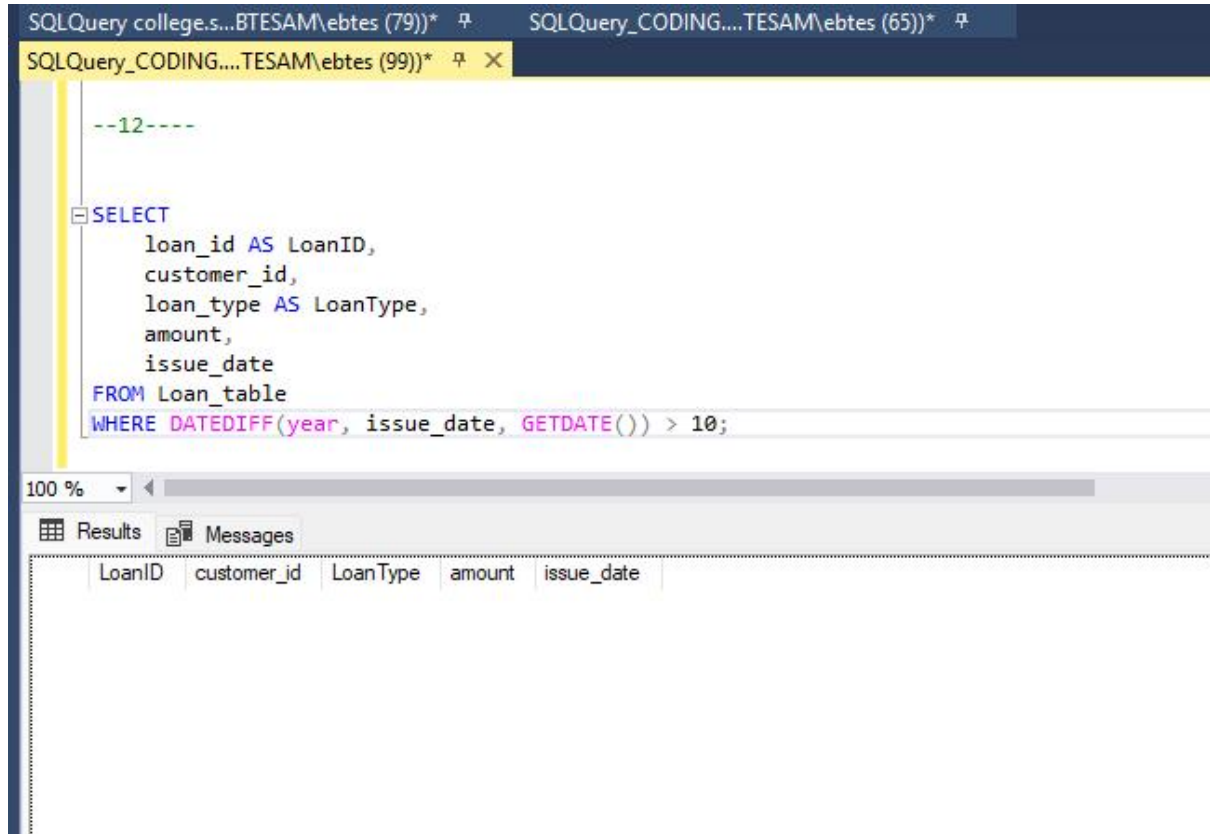
Results Messages

	TotalCustomers
1	4

11. Display customers with NULL phone numbers.



12. Display loans with duration greater than 10 years.



DML

13. Insert yourself as a new customer and open an account with balance 10000.

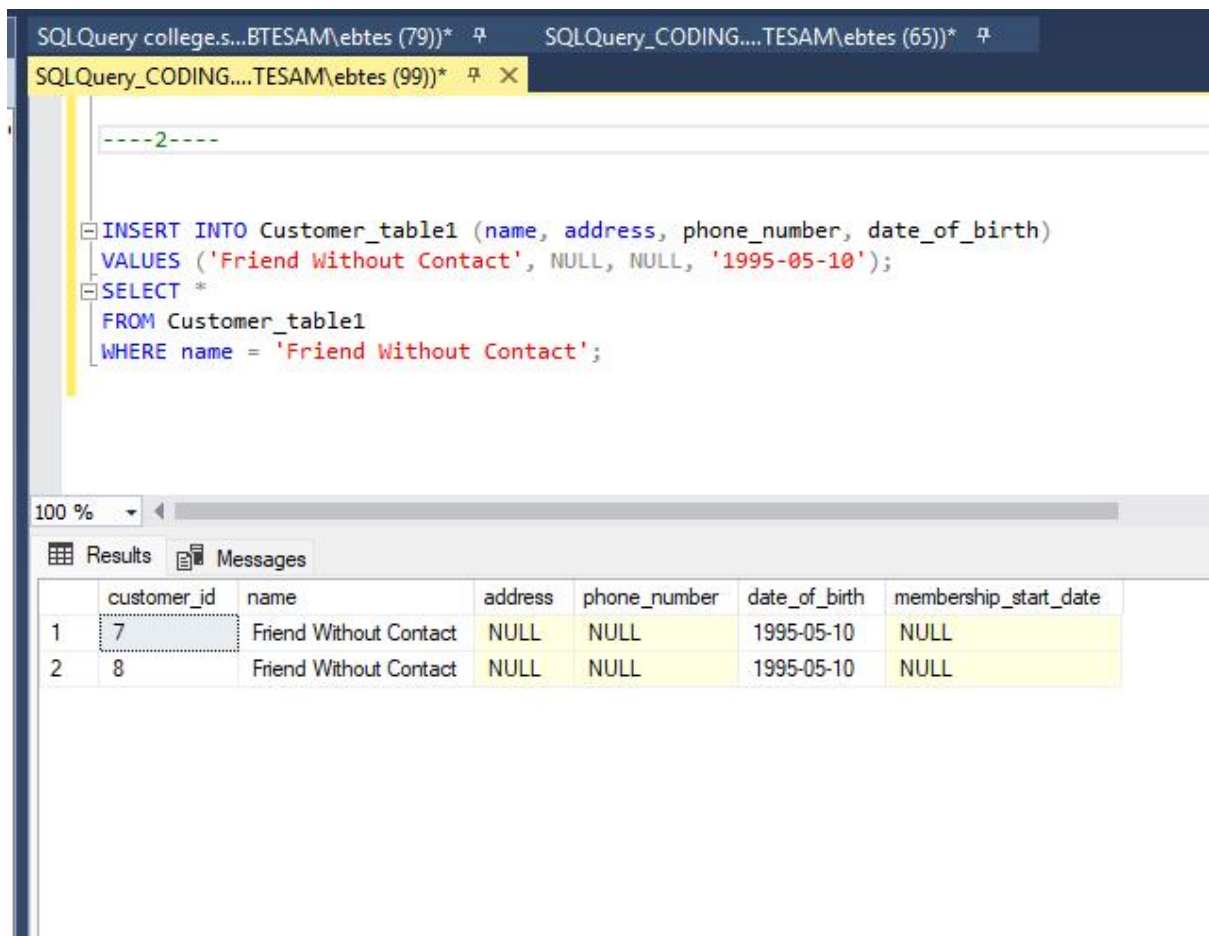
The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows a SQL query with the following code:

```
---1---  
  
INSERT INTO Customer_table1 (name, address, phone_number, date_of_birth)  
VALUES ('NASSER', '123 Some St, Muscat', '977000111', '1990-01-01');  
  
DECLARE @NewCustomerID INT;  
SET @NewCustomerID = SCOPE_IDENTITY();  
  
INSERT INTO Account_table (customer_id, balance, type, date_creation)  
VALUES (@NewCustomerID, 10000.00, 'savings', GETDATE());  
  
COMMIT;  
  
SELECT  
    c.customer_id,  
    c.name,  
    a.account_number,  
    a.balance,  
    a.date_creation
```

The bottom pane shows the 'Results' tab with a table containing 5 rows of data:

	CustomerID	CustomerName	AccountNumber	Balance	DateCreated
1	1	ebtesam	1	5000.00	2023-01-15
2	1	ebtesam	2	1200.00	2024-07-10
3	2	Fatma	3	8000.00	2022-03-20
4	3	Omar	4	300.00	2025-02-05
5	4	Amani	5	15000.00	2023-11-11

14. Insert another customer with NULL phone and address.



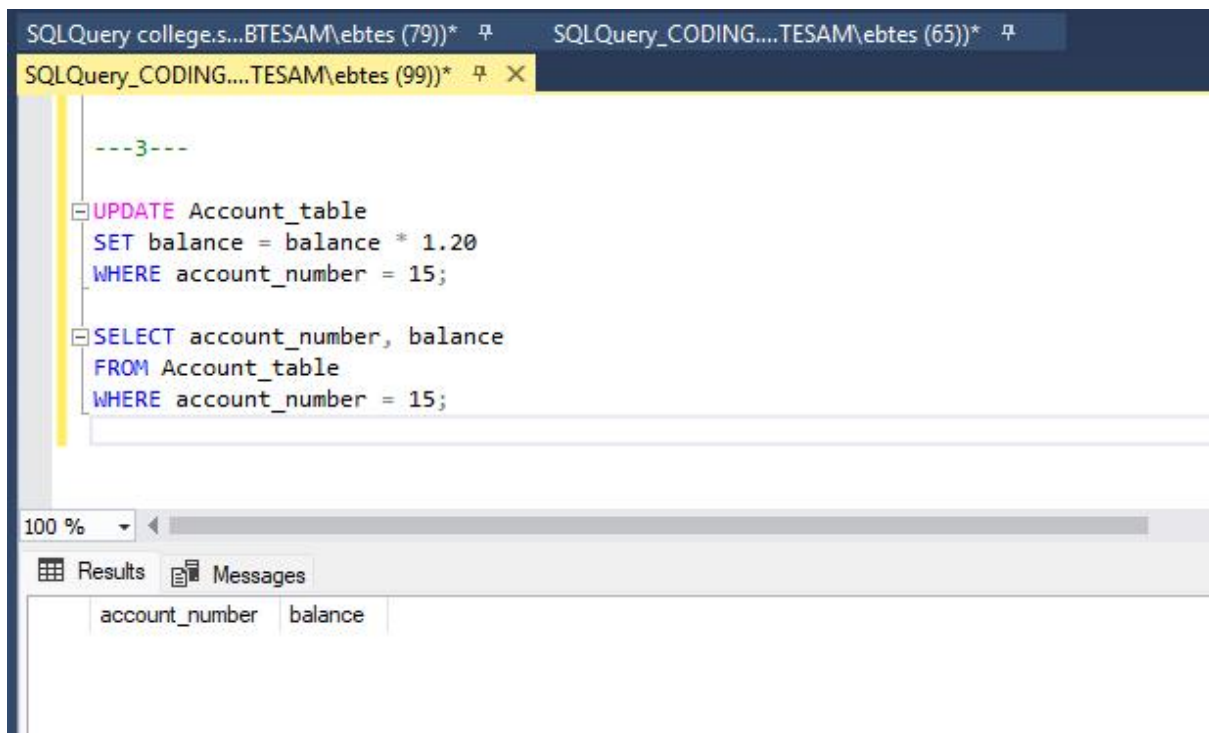
The screenshot shows the SQL Server Enterprise Manager interface. At the top, there are three query windows: 'SQLQuery college.s...BTESAM\ebtes (79))', 'SQLQuery_CODING....TESAM\ebtes (65))', and the active window 'SQLQuery_CODING....TESAM\ebtes (99))'. The active window contains the following SQL code:

```
-----2-----  
  
INSERT INTO Customer_table1 (name, address, phone_number, date_of_birth)  
VALUES ('Friend Without Contact', NULL, NULL, '1995-05-10');  
SELECT *  
FROM Customer_table1  
WHERE name = 'Friend Without Contact';
```

Below the query window, the 'Results' tab is selected, displaying a grid with two rows of data:

	customer_id	name	address	phone_number	date_of_birth	membership_start_date
1	7	Friend Without Contact	NULL	NULL	1995-05-10	NULL
2	8	Friend Without Contact	NULL	NULL	1995-05-10	NULL

15. Increase your account balance by 20%.

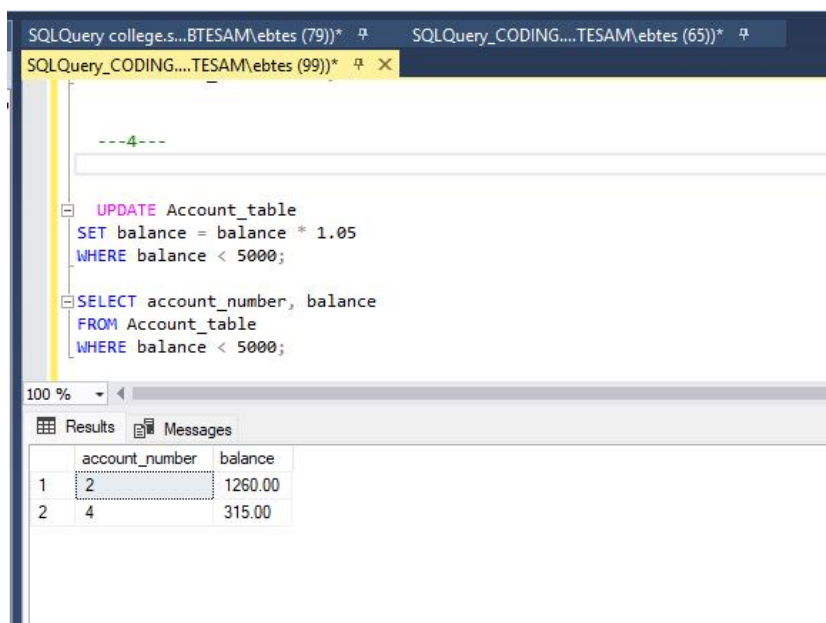


The screenshot shows a SQL Server Enterprise Manager interface. At the top, there are three tabs: 'SQLQuery college.s...BTESAM\ebtes (79))'*, 'SQLQuery_CODING....TESAM\ebtes (65))'*, and 'SQLQuery_CODING....TESAM\ebtes (99))'*. The active tab is 'SQLQuery_CODING....TESAM\ebtes (99))'*. The query window contains the following SQL code:

```
---3---  
  
UPDATE Account_table  
SET balance = balance * 1.20  
WHERE account_number = 15;  
  
SELECT account_number, balance  
FROM Account_table  
WHERE account_number = 15;
```

Below the query window, there is a 'Results' tab and a 'Messages' tab. The 'Results' tab is active, showing a table with two columns: 'account_number' and 'balance'.

16. Increase balance by 5% for accounts with balance less than 5000.



The screenshot shows a SQL Server Enterprise Manager interface. At the top, there are three tabs: 'SQLQuery college.s...BTESAM\ebtes (79))'*, 'SQLQuery_CODING....TESAM\ebtes (65))'*, and 'SQLQuery_CODING....TESAM\ebtes (99))'*. The active tab is 'SQLQuery_CODING....TESAM\ebtes (99))'*. The query window contains the following SQL code:

```
---4---  
  
UPDATE Account_table  
SET balance = balance * 1.05  
WHERE balance < 5000;  
  
SELECT account_number, balance  
FROM Account_table  
WHERE balance < 5000;
```

Below the query window, there is a 'Results' tab and a 'Messages' tab. The 'Results' tab is active, showing a table with two columns: 'account_number' and 'balance'. The table contains two rows of data:

	account_number	balance
1	2	1260.00
2	4	315.00

17. Update phone number to 'Not Provided' where phone is NULL.

The screenshot shows a SQL query window with the following code:

```
-----5-----  
UPDATE Customer_table1  
SET phone_number = 'Not Provided'  
WHERE phone_number IS NULL;  
  
SELECT customer_id, name, phone_number  
FROM Customer_table1  
WHERE phone_number = 'Not Provided';
```

Below the query window, the 'Results' tab is active, displaying a table with the following data:

	customer_id	name	phone_number
1	7	Friend Without Contact	Not Provided
2	8	Friend Without Contact	Not Provided

18. Delete closed accounts.

The screenshot shows a SQL query window with the following code:

```
---6---  
ALTER TABLE Account_table  
ADD status VARCHAR(20) NULL;  
  
UPDATE Account_table  
SET status = 'Closed'  
WHERE account_number = 123; -- use the real account number  
  
SELECT *  
FROM Account_table  
WHERE status = 'Closed';
```

Below the query window, the 'Results' tab is active, displaying a table with the following headers:

account_number	customer_id	balance	type	date_creation	status
----------------	-------------	---------	------	---------------	--------